**Department of Electrical Engineering**
**Indian Institute of Technology, Kanpur**

**EE 798V: Quantum Computation and Communication**

**Instructor:** Dr. Abhishek Gupta (email id: gkrabhi@iitk.ac.in), Dr. Washim Mondal (email id: wmondal@iitk.ac.in)

**Course TAs:** TBD

**Objectives:**

**Pre-requisite:** Linear Algebra, Probability Theory.

**Course Content:** The content for the course has been divided in several modules as follows:

| Content Outline of Individual Module | Tentative Number of Hours |
|---|---|
| Introduction, Classical vs Quantum Behavior, Applications | 1 |
| Linear Algebra Review, Vector space, operators, inner/outer product, Eigen values, Diagonalizability, Tensor products, trace and partial trace, Dirac Notation | 4 |
| Quantum postulates: Hilbert Space, time evolution, measurement, wavefunctions, Bloch sphere, POVM | 4 |
| Time Evolution of Pure State, Born Rule | 1 |
| State discrimination, Tensor Product, and Composite systems | 1 |
| Quantum Entanglement, Measurements (Partial, Destructive), bell states, quantum CNOT gate | 2 |
| Dense Coding, Teleportation, No-Cloning, | 2 |
| Density matrix, Ensemble, | 1 |
| Time Evolution, Purification, GHJW Theorem, Generalized measurements | 1 |
| Classical vs Quantum Computing, Quantum Gates and Universality | 1 |
| Classical Information Review, | 2 |
| Quantum Information: entropy, Joint entropy, relative entropy | 2 |
| Typical sequences, Shannon noiseless coding theorem and quantum version | 2 |
| Quantum Algorithms: Groover's, Simon's, Shor's, Deutsch–Jozsa Algorithm | 3 |
| Conditional gates, Representation of Unitary operators | 1 |
| EPR Paradox, Bell's Theorem, CHSH Inequality, Kochen-Specker Theorem | 2 |
| Generalized Measurement, Kraus Operator, | 2 |
| Quantum noise, quantum Channels, examples, decoherence Erasure, etc | 2 |
| Distance metrics: trace distance and fidelity | 1 |
| Holevo bound, HSW theorem, entanglement fidelity, communication over noisy quantum channels | 2 |
| Quantum communications, | 1 |
| Quantum Key Distribution (QKD), Continuous Variable QKD | 2 |

**References:**

1. Quantum Computation and Quantum Information by Nielsen & Chuang
2. Edward Witten "A Mini-Introduction To Information Theory", arXiv:1805.11965v5 4 Oct 2019
3. Lecture Notes MIT 8004 by B. Zwiebach
4. John Preskill "Lecture Notes on Quantum Computations. Available at http://theory.caltech.edu/~preskill/ph229/
5. Mark Wilde, "From Classical to Quantum Shannon Theory," http://arxiv.org/abs/1106.1445
6. Upmanyu Madhow, Digital Communications
7. Cover and Thomas, Elements of Information Theory

**Lecture:** Tuesday and Friday, 3.30 pm - 4.45 pm

**Grading scheme:**

- Class participation: 20% (Attendance + discussion + scribing)
- Assignments: 20% (4 - 6 assignments throughout the semester)
- Exams: 20% (Mid-sem) + 20% (End-sem)
- Course project: 20%